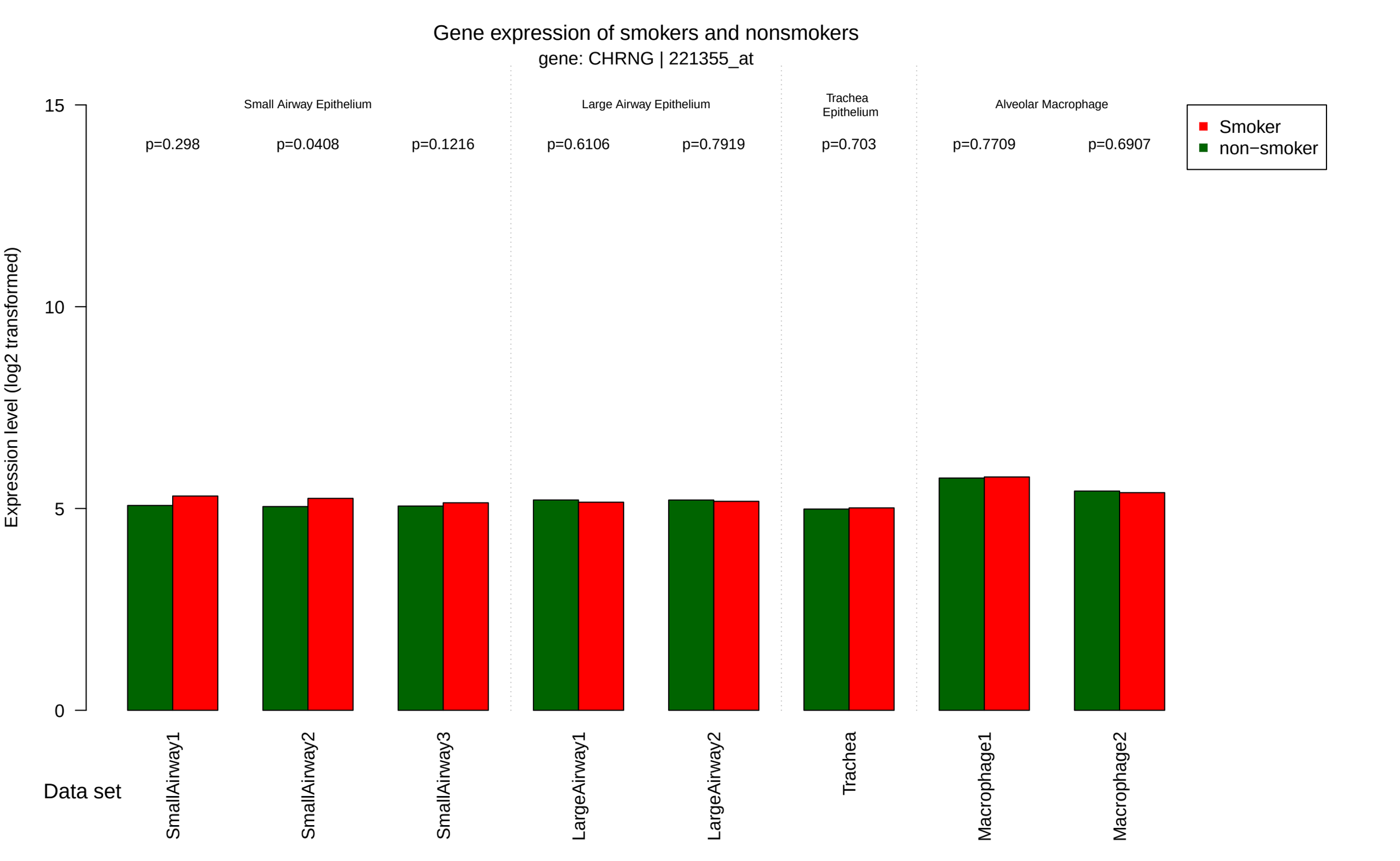
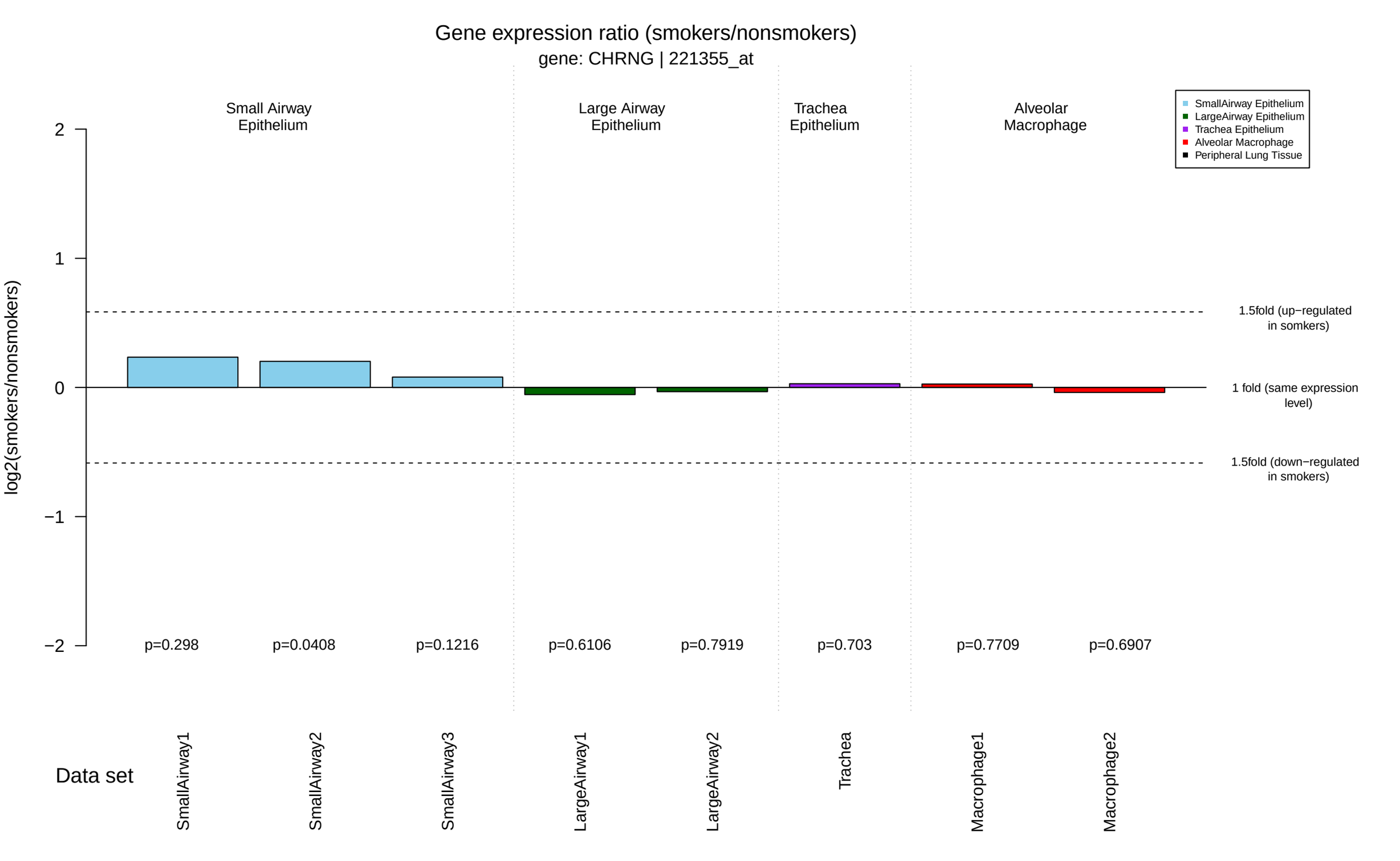
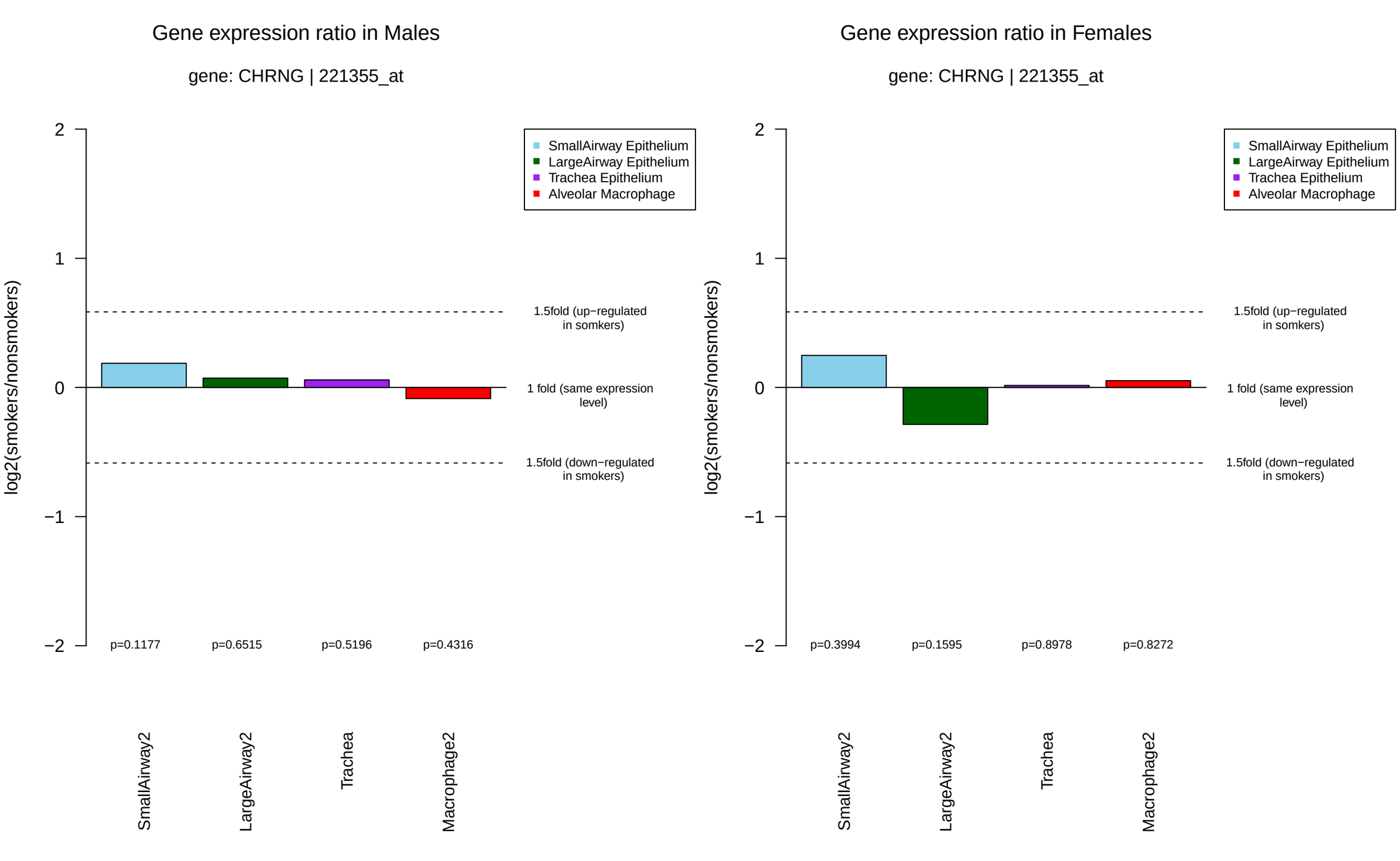
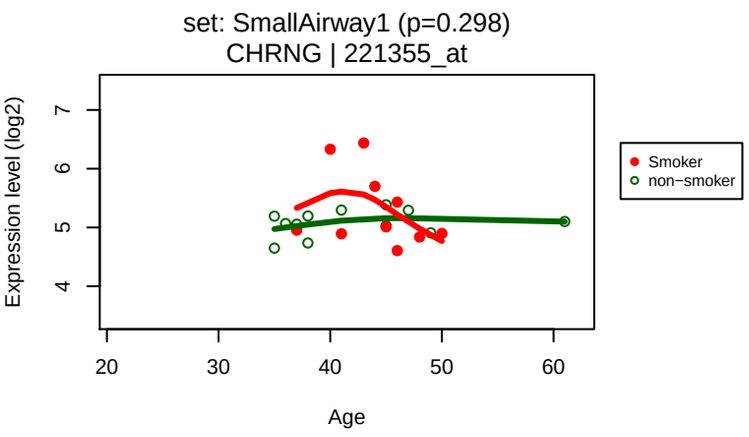
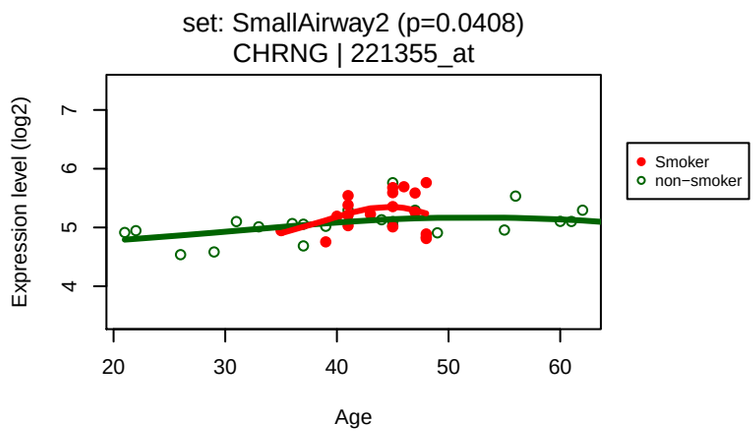
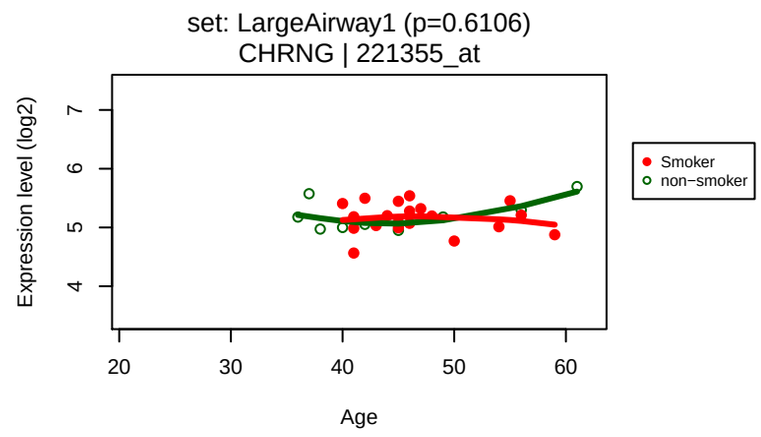

Gene expression of smokers and nonsmokers
gene: CHRNG | 221355_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.298
p=0.0408
p=0.1216
p=0.6106
p=0.7919
p=0.703
p=0.7709
p=0.6907
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

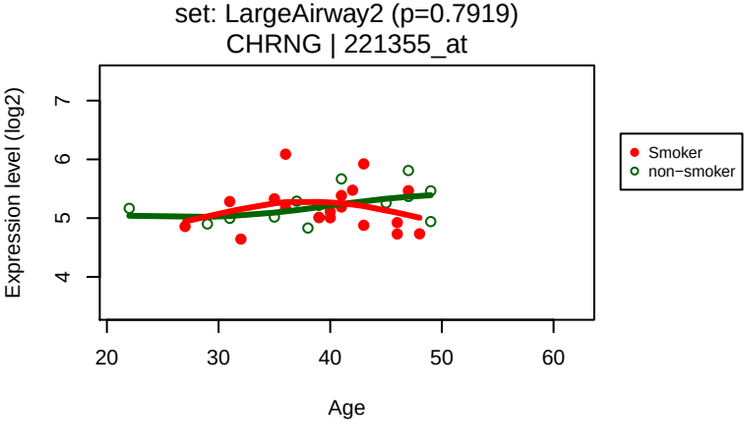

Gene expression ratio (smokers/nonsmokers)
gene: CHRNG | 221355_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.298
p=0.0408
p=0.1216
p=0.6106
p=0.7919
p=0.703
p=0.7709
p=0.6907
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

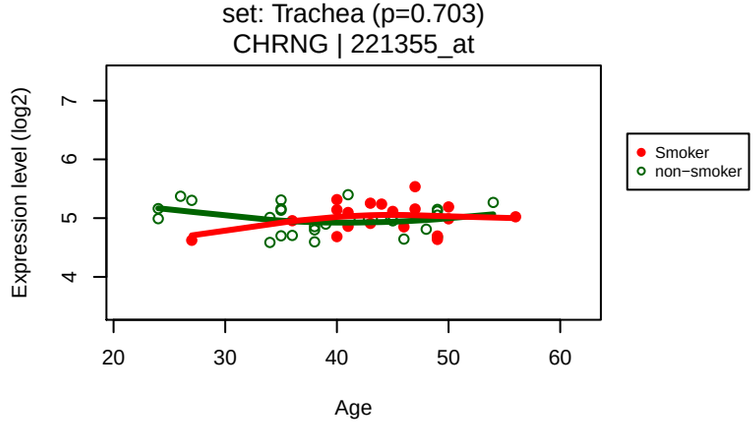

Gene expression ratio in Males
gene: CHRNG | 221355_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1177
p=0.6515
p=0.5196
p=0.4316
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNG | 221355_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3994
p=0.1595
p=0.8978
p=0.8272
SmallAirway2
LargeAirway2
Trachea
Macrophage2

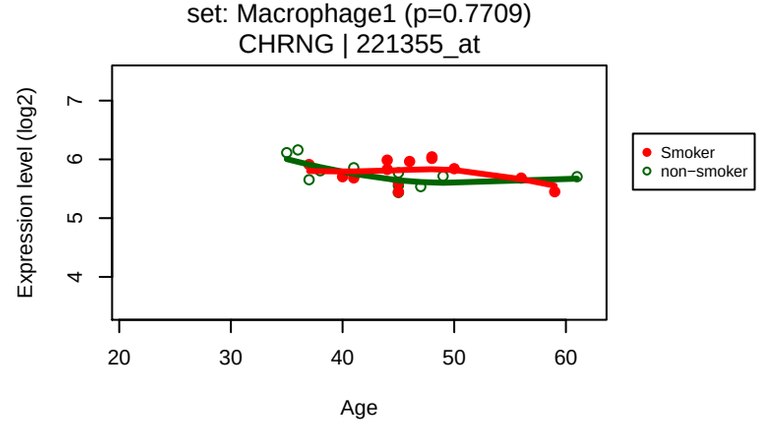

set: SmallAirway1 (p=0.298)
CHRNG | 221355_at
Expression level (log2)
Age
Smoker
non−smoker

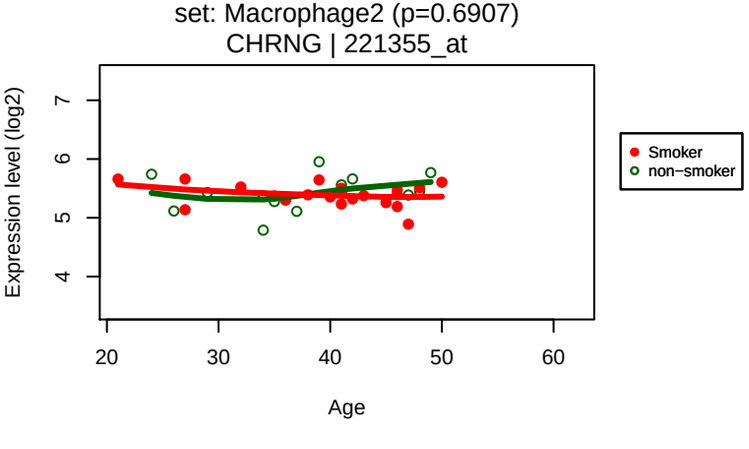

set: SmallAirway2 (p=0.0408)
CHRNG | 221355_at
Expression level (log2)
Age
Smoker
non−smoker

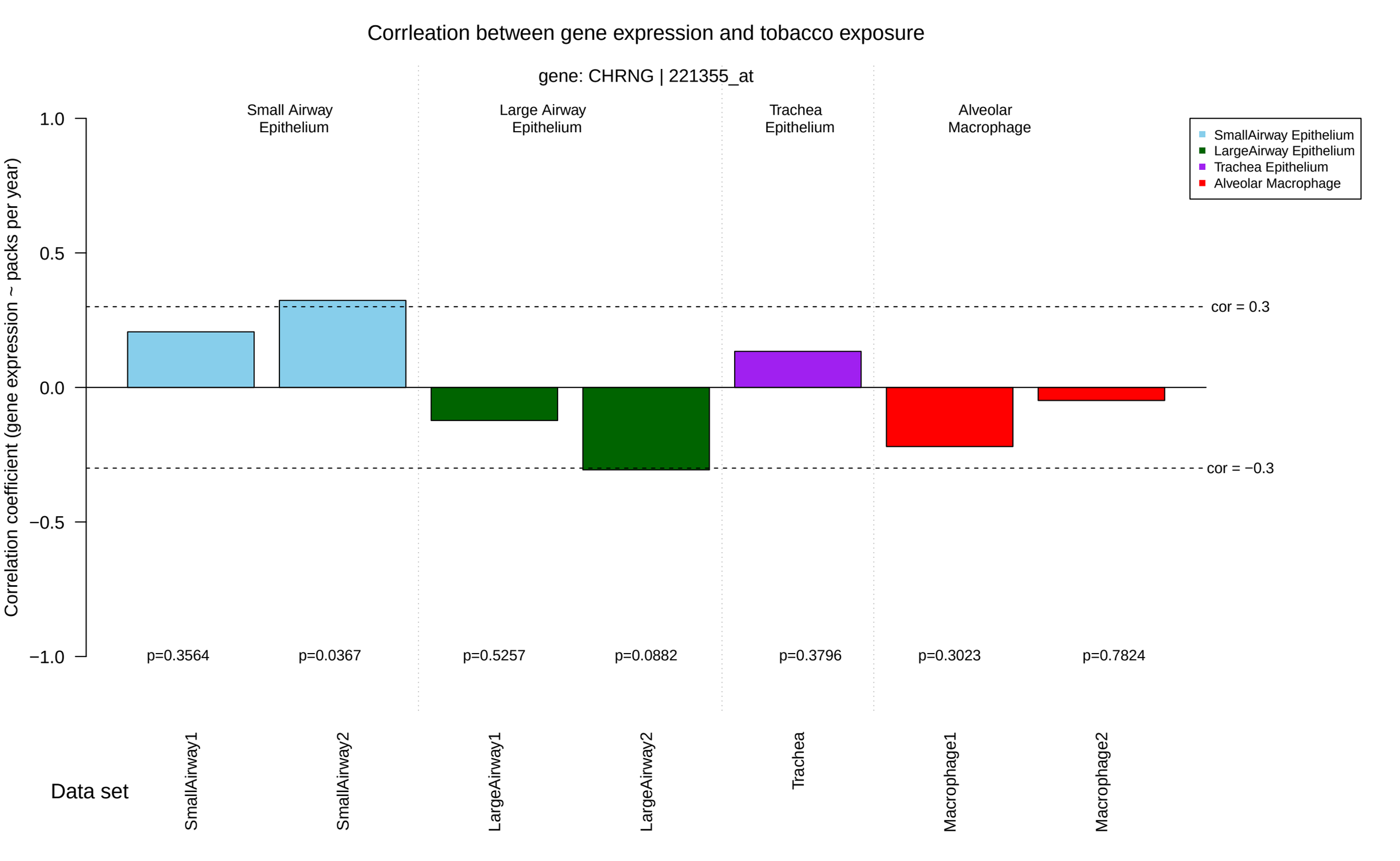

set: LargeAirway1 (p=0.6106)
CHRNG | 221355_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7919)
CHRNG | 221355_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.703)
CHRNG | 221355_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7709)
CHRNG | 221355_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6907)
CHRNG | 221355_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNG | 221355_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3564
p=0.0367
p=0.5257
p=0.0882
p=0.3796
p=0.3023
p=0.7824
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2